

Truly-Operational Quantum-Hybrid PINN for Cryogenic Hydrogen Infrastructure Integrity

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Abstract — This paper introduces a novel simulation framework, **Quantum-Hybrid PINN**, designed for the safety certification of liquid hydrogen (LH2) infrastructures. By embedding Navier-Stokes equations and NIST-validated thermodynamic properties directly into a deep learning architecture using PyTorch Autograd, we eliminate physical hallucinations common in traditional AI models. We propose a sequential **G0-G5 certification protocol** that bridges the gap between B-Rep CAD modeling and automated physical audit. Experimental results on 1,250 m³ storage tanks and SAE J2601-2 refueling systems demonstrate physical residuals below 10⁻⁷, validated through a high-performance Fortran/Python hybrid solver.

1. Introduction

The global energy transition relies heavily on the large-scale deployment of liquid hydrogen. However, cryogenic fluid dynamics present extreme challenges for traditional CFD due to phase change and non-ideal gas behavior. This work presents a “Physics-First” digital twin that ensures mathematical integrity through automated residual monitoring.

2. Methodology: The G0-G5 Protocol

The framework is structured around five mandatory validation gates:

- **G0/G1**: Topological manifold validation of STEP AP242 geometries.
- **G2**: High-fidelity tetrahedral mesh generation via Gmsh.
- **G3/G4**: Constraint of the PINN loss function by Navier-Stokes and NIST REFPROP.
- **G5**: Certification of Autograd residuals under industrial tolerances.

3. Experimental Results

For a 1,250 m³ LH2 sphere at 20.28 K, the model achieved a credibility score of 99.8%. The hybrid solver, utilizing Fortran 90 kernels with OpenMP parallelism, reduced computation time by 90% compared to pure Python implementations while maintaining sub-millimeter spatial resolution.

Metric	Value	Status
Mass Residual	1.15×10^{-7}	Certified
Momentum Residual	3.42×10^{-7}	Certified

Energy Residual	5.89×10^{-7}	Certified
Credibility Score	99.8%	Validated

4. Discussion: Hybrid FNO/PINN Synergy

The integration of Fourier Neural Operators with Physics-Informed Neural Networks addresses the fundamental “spectral bias” of traditional deep learning models. While standard PINNs often struggle to capture high-frequency gradients in transient cryogenic flows, the FNO component provides a robust global solution in the frequency domain, reducing initial Mean Squared Error (MSE) by 40-60%.

The model’s robustness against pressure shocks and discontinuities is achieved through a multi-scale decomposition strategy. The FNO ensures global stability, preventing numerical divergence during rapid 35 MPa refueling transients, while the PINN component, powered by PyTorch Autograd, refines local gradients without the constraints of a fixed mesh. This hybrid loss function, dynamically weighted to penalize physical conservation violations, ensures that residuals remain below the 10^{-7} threshold even in the presence of steep thermal fronts and boil-off phenomena.

5. Conclusion

The Quantum-Hybrid PINN platform establishes a new standard for industrial simulation. By combining the speed of neural inference with the rigor of classical physics, it provides a scalable solution for the certification of critical hydrogen infrastructure.